Exploratory Factor Analysis and Model Development

Jinsong Chen

Table of contents

1 Define the exploratory measurement problem **3**
1.1 The common-factor model is not restriction-free 4
1.2 A twelve-indicator content record 4
1.3 Audit before interpreting correlations 5

2 Choose the matrix and extraction target **7**
2.1 Covariance and correlation answer different scaling questions 7
2.2 PCA: a decomposition of total observed variance 9
2.3 Principal-axis factoring: estimate the diagonal contribution 10
2.4 Minimum residual and maximum likelihood extraction 11

3 Determine a plausible factor count **13**
3.1 Underfactoring and overfactoring have different costs 13
3.2 Specify the matrix whose eigenvalues are being read 13
3.3 Scree and the eigenvalue-greater-than-one heuristic 14
3.4 Parallel analysis with a reproducible null comparison 14
3.5 A wider family of retention rules 17

4 Estimate neighboring dimensional solutions **19**
4.1 Count effective parameters, not every printed loading 19
4.2 Fit one through five; inspect two, three, and four closely 20
4.3 An exact-fit test is one piece of evidence 21

5 Resolve the orientation by rotation **22**
5.1 What rotation changes, and what it preserves 22
5.2 Orthogonal and oblique simple structure 23
5.3 Why the oblique solution is primary here 24

6 Read an oblique factor system **25**
6.1 Pattern coefficients and structure correlations . . . 25
6.2 Communality is a quadratic form . . . 26
6.3 Reconstruct a pair, then the whole matrix . . . 27
6.4 What is uncertain about this interpretation? . . . 28

7 Check reconstruction and analytic sensitivity **28**
7.1 Examine local residuals, not only an average . . . 28
7.2 KMO and Bartlett answer preliminary questions . . . 30
7.3 Change one analytic choice at a time . . . 31

8 Summary and reading bridge **32**

Exercises **33**

Appendix A: SLE data dictionary and sample manifest **34**
Fields and common coding . . . 34
Item codebook . . . 35
Sample manifest . . . 36

Appendix B: Spectral and rotation details **36**
Why a truncated eigendecomposition is a component solution . . . 36
Other retention summaries are not interchangeable votes . . . 37
Two explicit oblique criteria . . . 37
Factor-score indeterminacy in one equation . . . 38

References **38**

Chapter box

Five themes.
The direction of work reverses. The preceding chapter built a covariance matrix from declared parameters. Here a sample correlation matrix comes first, and no simple model reproduces it exactly.
Every diagnostic compares against a reference. Extraction, retention and rotation each state what the data are being held up against. Naming the reference is what makes a result interpretable, and what makes two procedures with the same name different.
Coordinates are chosen, not found. Rotation changes the loading display without changing the fitted covariance, so pattern coefficients, structure correlations and factor correlations are three different quantities that must be read together.
Discrepancy has a location. A residual correlation matrix says where a retained solution strains; an average residual cannot.
A development sample cannot confirm what it was used to develop. The analysis produces a candidate whose restrictions are stated explicitly, and the evidence for it has to come from records not yet inspected.
Prerequisites. Sample covariance and correlation, elementary matrix algebra, and the common-factor model, communalities, uniquenesses and coordinate indeterminacies developed in the preceding chapter. Basic R helps reproduction; the printed matrices, equations and output support the reasoning without running it.
Record. Sample A contains 360 complete, simulated continuous records on twelve Student Learning Experiences indicators. No observations concern actual students. Samples B and C are not analysed here.
Scope. We develop a candidate measurement model on Sample A and stop there. No confirmation sample is opened, no restricted model is estimated, and no fit index is interpreted as evidence for a construct. Indicators are treated as continuous throughout Part Two. This is model development, not instrument validation or individual scoring.

1 Define the exploratory measurement problem

Chapter 1 began with known loadings and factor covariances and worked forward to an implied covariance matrix. Here the direction of work changes. We observe a sample of indicator values, calculate their associations, and ask which common-factor representation is defensible. The observed correlations fluctuate from sample to sample. Their pattern does not arrive with a factor count, a preferred rotation, or substantive names attached. Those have to be developed and documented.

Exploratory factor analysis, or **EFA**, leaves most loading restrictions open while seeking a comparatively low-dimensional representation of shared covariation. It is not a procedure with no assumptions. Selecting the indicators, defining the population, choosing an association matrix, restricting residual covariance, considering a range of dimensions, and choosing rotation criteria all shape what the analysis can reveal. A carefully designed exploratory study therefore has substantial work to do before an extraction routine runs.

The distinction between a provisional representation and a confirmed restriction is important from the outset. Suppose an EFA suggests that an indicator of adaptive planning also relates to strategy adjustment. That can be a useful discovery. It is not evidence that a cross-loading specified after seeing these observations was predicted before them. The later CFA can evaluate the resulting restriction on unused data, but changing the name of the fitting function does not change the history of the decision.

1.1 The common-factor model is not restriction-free

For person or observational unit i, retain Chapter 1's notation:

$$\mathbf{y}_i = \nu + \Lambda\eta_i + \epsilon_i, \qquad \Sigma = \Lambda\Phi\Lambda^{\mathsf{T}} + \Theta. \tag{1}$$

There are p indicators and m factors. Thus Λ is $p \times m$, Φ is $m \times m$, and Θ and Σ are $p \times p$. The factor-residual covariance is zero. Classical EFA initially makes Θ diagonal: remaining indicator-specific variation does not covary across indicators. That restriction can be wrong even when most loadings are freely estimated.

The loading pattern is initially open, but its dimension is not. A two-factor and a three-factor EFA are different covariance models. Scale restrictions are also necessary. One convenient extraction representation uses unit-variance, uncorrelated factors; a subsequent oblique transformation expresses the same common covariance using correlated coordinates. Orthogonality at this computational stage does not establish independence of substantively named attributes.

The latent intercept α equals the factor mean in this model without structural paths or separate predictors. It remains an intercept symbol throughout Part Two. Likewise Φ denotes total factor covariance, not residual covariance and not the later structural-disturbance matrix. These distinctions prevent the same letter from silently changing meaning between EFA, CFA, and SEM.

1.2 A twelve-indicator content record

The sustained example concerns **Student Learning Experiences (SLE)**. The proposed domains are Planning, Strategy, and Belonging. These names organize the starting content questions; they do not determine how many factors Sample A will support. All twelve variables are explicitly simulated continuous process indices. Their units are index points, not response categories, percentages, or counts. The statements below clarify intended content, but they are not presented as an administered and validated questionnaire.

Table 1
Proposed content before factor analysis. All indices are positively directed.

item	domain	wording
P1	Planning	I establish a study plan before beginning a task.
P2	Planning	I identify which study tasks deserve attention first.
P3	Planning	I compare my progress with the plan I made.
P4	Planning	I revise my plan when a study method is not working.
S1	Strategy	I change my learning strategy when understanding is weak.
S2	Strategy	I select a learning method suited to the task.
S3	Strategy	I revise my explanation after checking my understanding.
S4	Strategy	I use discussion with others to improve my learning method.
B1	Belonging	I feel accepted by other students.
B2	Belonging	I feel able to join other students in learning activities.
B3	Belonging	I feel able to express my views in class.
B4	Belonging	I feel connected to the university as a whole.

P4 concerns changing a plan when a learning method fails. S4 concerns using discussion with others to improve a learning method. Their content already suggests possible overlap between domains. A cross-loading would not automatically make either indicator defective. Conversely, clear wording does not guarantee that its covariance pattern will match the intended domain.

The twelve columns have different raw dispersions. All are coded so that larger values mean more of the stated attribute, and none is reverse-scored. The simulation imposes neither missing observations nor response limits. Its unbounded normal records are a deliberate teaching design, not a claim that actual student experiences are normally distributed or measured without limits.

Download sle_continuous.csv, the only SLE data file needed for the examples. The SLE data appendix contains the complete codebook and sample manifest. Its `sample` column identifies three independent 360-case samples, not three groups to pool. Select A before inspecting indicators or calculating statistics in this chapter. Population parameters remain in the documented generator, separate from the learner-facing analysis: knowing the recipe is not a substitute for reasoning from Sample A. No empirical summary, correlation, factor solution, or fit statistic from B or C has been used to make the decisions reported here.

1.3 Audit before interpreting correlations

Data auditing is part of defining the statistical input. An unintended reverse code can change signs. A nearly constant indicator can make a correlation unstable. Duplicate records alter the apparent amount of information. Pairwise deletion can produce different effective samples for different correlations, and sometimes a matrix that is not positive definite. None of these problems is repaired merely by rotating a loading matrix until it looks recognizable.

Table 2
Sample A audit, in raw process-index points; skewness is the third standardized central moment.

Item	Mean	SD	Min	Max	Skew	Missing
P1	50.27	7.93	25.76	73.42	-0.05	0
P2	52.76	10.20	25.66	84.47	0.26	0
P3	49.02	12.12	8.28	86.27	-0.08	0
P4	51.44	9.36	25.88	75.07	0.05	0
S1	48.59	10.17	21.39	75.02	0.00	0
S2	50.28	7.81	26.68	71.48	-0.12	0
S3	51.28	10.91	24.32	79.54	0.02	0
S4	48.09	8.70	22.76	73.89	-0.17	0
B1	51.26	10.07	18.59	82.73	0.02	0
B2	49.62	12.60	17.60	96.23	-0.08	0
B3	50.09	8.25	28.04	74.26	0.07	0
B4	51.50	11.30	22.14	92.91	0.16	0

There are 360 distinct rows, no missing values, and positive variance in every indicator. The minimum eigenvalue of the Pearson matrix is 0.223, so its inverse exists without matrix repair. This is a useful computational check, not evidence that the factor model is true. Normal-theory ML also needs an appropriate sampling and distributional interpretation. The simulation supplies independent normal factor/residual draws; a real study would have to justify or investigate those assumptions.

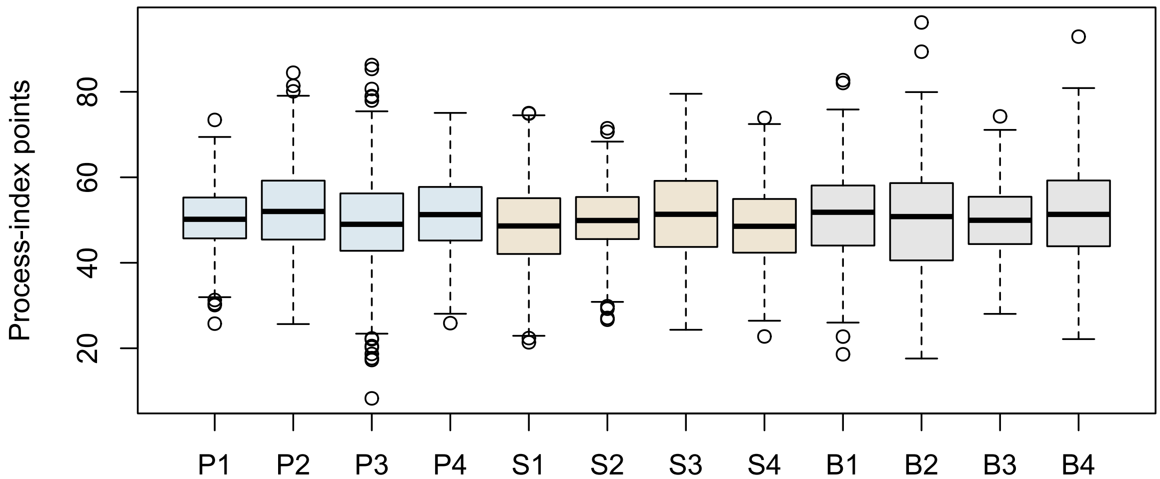

Figure 1: Raw distributions in Sample A. Different dispersions motivate an explicit scaling choice.

In an actual application, an unusual observation deserves examination of provenance and measurement, not automatic removal because it makes normality less convenient. A miscoded value, a genuine extreme case, and a member of another target population call for different actions. Any exclusion changes the estimand's population or the analyzed record and must be described. Here no item or case is removed. That keeps the same twelve-variable target available for later confirmatory comparisons.

2 Choose the matrix and extraction target

2.1 Covariance and correlation answer different scaling questions

Let $\mathbf{S}$ denote the ordinary sample covariance matrix computed with denominator $n-1$, and let $\mathbf{D}_S = \text{diag}(s_{11}, \dots, s_{pp})$. The Pearson correlation matrix is

$$\mathbf{R} = \mathbf{D}_S^{-1/2}\mathbf{S}\mathbf{D}_S^{-1/2}. \tag{2}$$

This operation divides each covariance by the two sample standard deviations. It does not remove dependencies, correct measurement error, or make a nonnormal distribution normal. It changes the units in which the association structure is represented. Covariance-scale analysis preserves raw variances; correlation-scale analysis gives each observed variable a unit diagonal.

Our primary analysis uses **R**. The rationale is that the process-index dispersions are not intended to set the relative weight of the content domains. Raw **S**, means, and standard deviations are retained, however. Later CFA fits raw observations, and an explicit scale transformation connects those fits with this standardized EFA. No variable is replaced by an ordinal score or a new construct along the way.

```r
# The development sample is simulated from the declared population, so this
# runs without any data file. Appendix A gives the population and seeds.
X <- sle_simulate("development")$x
S <- cov(X)
R <- cor(X)
all.equal(R, cov2cor(S))
```

The last line returns `TRUE` for these complete observations. It checks the transformation, not the suitability of Pearson correlation for every dataset. Numeric codes for categories would require a separate measurement treatment; that topic belongs to other courses or later series. Categorical group labels and coded predictors remain possible in later multigroup or MIMIC work. This brief boundary introduces no categorical-estimator track here.

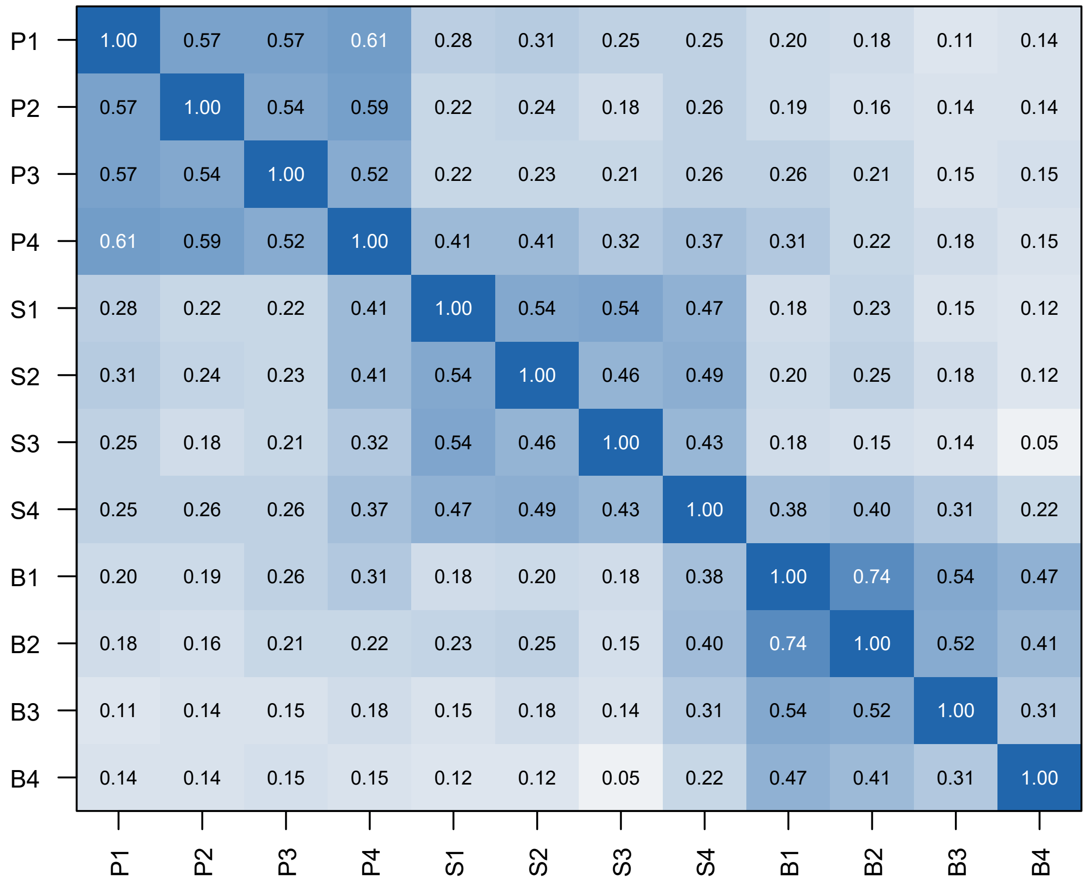

Figure 2: Sample A Pearson correlations, with a fixed scale from -1 to 1. The ordering follows the content map, not a fitted cluster algorithm.

The broad blocks are suggestive, not conclusive. A correlation between a Planning and a Strategy indicator might arise from correlated factors, a cross-loading, residual dependence, or some combination. The same visible association can therefore be represented by different restrictions. The task is not to give each colored block a name and stop; it is to evaluate a coherent factor representation of the matrix.

2.2 PCA: a decomposition of total observed variance

For a symmetric positive-definite correlation matrix,

$$\mathbf{R} = \mathbf{V}\mathbf{D}\mathbf{V}^{\mathsf{T}}, \qquad \mathbf{V}^{\mathsf{T}}\mathbf{V} = \mathbf{I}, \tag{3}$$

where the columns of $\mathbf{V}$ are orthonormal eigenvectors and $\mathbf{D} = \mathrm{diag}(d_1, \ldots, d_p)$ contains ordered eigenvalues. A rank-m PCA reconstruction uses

$$\mathbf{A}_m = \mathbf{V}_m \mathbf{D}_m^{1/2}, \qquad \widehat{\mathbf{R}}_{\mathrm{PCA},m} = \mathbf{A}_m \mathbf{A}_m^{\mathsf{T}}. \tag{4}$$

The full decomposition is exact; a truncated one generally is not. PCA chooses directions of high observed variance. Component scores are weighted functions of observed variables, not unobserved random variables whose common contribution is separated from item residuals. Calling $\mathbf{A}_m$ a loading matrix in software does not change that distinction.

The first three Sample A component eigenvalues sum to 7.809, or 65.1% of the twelve units of standardized observed variance. This is a PCA summary. It is not the common variance attributed to three latent factors by the ML model. The two summaries can differ because their objectives differ, even when their broad dimensional patterns look similar.

PCA may be appropriate when compact prediction, compression, or a descriptive coordinate system is the objective. The problem is not using PCA; it is interpreting its output as if a common-factor measurement model had been estimated. A method should be judged against the task it actually solves.

2.3 Principal-axis factoring: estimate the diagonal contribution

Principal-axis factoring, or PAF, works with a reduced correlation matrix. At iteration t, replace the unit diagonal by estimated communalities:

$$\mathbf{R}_c^{(t)} = \mathbf{R} - \mathrm{diag}(1 - \hat{\mathbf{h}}^{2(t)}). \tag{5}$$

Decompose this matrix, use its leading positive eigenvalues to form a provisional loading matrix, and update the communalities from its row sums of squared loadings in an orthogonal representation. Repeat until a declared rule indicates sufficiently small changes. The initial diagonal, handling of nonpositive roots, and stopping rule are implementation choices, not properties of factor analysis itself.

One common initialization is the squared multiple correlation of each variable with the remaining variables. For nonsingular $\mathbf{R}$,

$$\hat{h}_{j,0}^2 = 1 - \frac{1}{(\mathbf{R}^{-1})_{jj}}. \tag{6}$$

These are regression summaries of the observed matrix, not known common variances. They can underestimate some true communalities and can be unstable when predictors are highly redundant. They provide a starting point, not an answer to the factor-count question.

Table 3

A single three-factor PAF-style spectral update from an SMC diagonal. This is an illustration, not the primary ML fit or a converged PAF solution.

Item	SMC	First update
P1	0.496	0.576
P2	0.458	0.535
P3	0.424	0.494
P4	0.550	0.616
S1	0.455	0.545
S2	0.409	0.483
S3	0.369	0.439
S4	0.410	0.472
B1	0.637	0.720
B2	0.599	0.665
B3	0.335	0.383
B4	0.243	0.271

For P1, the SMC start is 0.496 and the first updated communality is 0.576. Repeating that update would change the reduced diagonal again. This explains why a PAF root is not automatically the same as a root of the original unit-diagonal correlation matrix.

2.4 Minimum residual and maximum likelihood extraction

Minimum-residual extraction seeks small discrepancies between observed and reproduced off-diagonal correlations. A representative objective is

$$F_{\text{minres}} = \sum_{j<k} \left[r_{jk} - (\Lambda\Phi\Lambda^{\mathsf{T}})_{jk} \right]^2 . \tag{7}$$

The diagonal residual variances then complete the observed variances. This criterion gives the same squared-error weight to each displayed off-diagonal discrepancy. Software algorithms can differ in numerical details; the objective, convergence criteria, and resulting covariance still need to be checked. PAF and minres are both common-factor approaches, but they are not names for the same algorithm.

The primary method here is **maximum-likelihood common-factor extraction**. For an observed covariance matrix $\mathbf{S}$ and a positive-definite model-implied covariance $\Sigma(\vartheta)$, the normal-theory discrepancy can be written

$$F_{\mathrm{ML}}(\vartheta) = \log|\Sigma(\vartheta)| + \mathrm{tr}\big[\mathbf{S}\Sigma(\vartheta)^{-1}\big] - \log|\mathbf{S}| - p. \tag{8}$$

Here ϑ collects free parameters; it is not the residual matrix Θ. The determinant terms measure covariance volume, while the trace term measures discrepancy relative to the model's covariance geometry. The two sample-only terms set the discrepancy to zero for exact reproduction. Minimization chooses parameter values under the model's restrictions, rather than choosing a covariance matrix independently of them.

The same expression applies to our correlation-scale analysis with $\mathbf{R}$ in place of $\mathbf{S}$. Under freely rescalable loadings and residual variances, transforming observed units has a corresponding parameter transformation. This does not mean every arbitrary constraint is invariant to units. Equal unstandardized loadings, for example, must be re-expressed when the indicators change scale.

Normal ML supplies a distributional framework for standard errors and test statistics when its assumptions and regularity conditions hold. Independent observations, a suitable population sampling interpretation, identification, and admissible covariance matrices matter. Good-looking loadings cannot supply any of these by themselves. Numerical convergence only says that the optimizer met a stopping rule; it does not show that the fitted representation is adequate or that the solution is uniquely identified.

Method	Immediate target	What the output does not establish
PCA	Low-rank reconstruction of total observed variance	A latent measurement decomposition
PAF	Iterative reduced-matrix factor representation	A universal diagonal start or stopping tolerance
Minres	Small off-diagonal squared residuals	Normal-likelihood inference without additional work
ML factor analysis	Best normal likelihood within the factor model	Substantive truth, validity, or acceptable fit merely from convergence

The maintained sequence is therefore Pearson correlations, ML extraction, and oblimin rotation. Minres is a bounded sensitivity comparison on the same matrix. The later CFA uses the same continuous ML family; it is not a transition from one response model to another.

3 Determine a plausible factor count

3.1 Underfactoring and overfactoring have different costs

Too few factors force distinctions into an inadequate space. Indicators of an omitted dimension can acquire misleading loadings on the retained dimensions, factor correlations can shift, and structured residuals can remain. Too many factors can divide one broad pattern into weak or unstable directions, make residual variances approach a boundary, and encourage names for sample-specific features. Adding a dimension is not free just because numerical reconstruction improves.

Retention is therefore a question about the representation to carry forward, not a race to maximize variance accounted for. We consider one through five factors before examining the results. Three is a content-based expectation, not a required outcome of the analysis. The full range is retained in the record even when neighboring models give inconvenient results.

3.2 Specify the matrix whose eigenvalues are being read

A spectral eigenvalue satisfies

$$\mathbf{R}\mathbf{v}_k = d_k \mathbf{v}_k. \tag{9}$$

Its meaning depends on the matrix. Eigenvalues of the full Pearson matrix include all standardized observed variance and sum to p. Eigenvalues of an SMC-reduced matrix use a different diagonal and need not all be positive. An estimated common covariance has yet another spectrum. A printed column headed "eigenvalue" must therefore be interpreted alongside the software's construction of that column.

The post-rotation column statistic

$$SS_k = \sum_{j=1}^{p} \lambda_{jk}^2 \tag{10}$$

is an **SS loading**, not generally a spectral eigenvalue. In suitable unrotated orthogonal decompositions these quantities coincide; after rotation the column allocation can change while common covariance stays fixed. With correlated factors, pattern-column SS values do not provide a unique additive partition of common variance. Chapter 1's rotated two-factor example already demonstrated the simpler orthogonal version of this distinction.

3.3 Scree and the eigenvalue-greater-than-one heuristic

The familiar Kaiser–Guttman heuristic retains roots above one in a unit-diagonal correlation analysis. It compares a component's variance with one observed standardized variable's variance. Sampling variation alone can create roots above one, so this is neither a necessary condition for a useful common factor nor a reliable stand-alone count. Applying the same threshold to a reduced matrix changes the comparison and should not be treated as an interchangeable implementation.

A scree plot instead asks where large decreases give way to a flatter sequence. It can reveal a distinct elbow, multiple plausible elbows, or no clear break. The method involves judgment: a visually appealing elbow does not supply a sampling reference. Full- and reduced-matrix plots need not agree, even when they do in the present example.

In Sample A, the first three full-matrix roots are 4.380, 1.942, 1.487, followed by 0.705. That drop makes three a plausible candidate. It becomes more informative when compared with what equally sized data containing no cross-variable dependence would produce.

3.4 Parallel analysis with a reproducible null comparison

Parallel analysis compares ordered observed eigenvalues with a matched reference distribution (Buja & Eyuboglu, 1992; Horn, 1965). For dimension k, our comparison is

$$d_k^{\text{observed}} > Q_{.95}\{d_{k,1}^{\text{null}}, \ldots, d_{k,1000}^{\text{null}}\}. \tag{11}$$

The reference is the 95th percentile, not the mean. We independently permute the 360 entries of each observed column **without replacement**. This preserves each empirical marginal distribution exactly while breaking the original matching of observations between columns. Every null dataset still has 360 rows and twelve variables.

Two comparisons are kept separate. Component PA uses eigenvalues of the full Pearson matrix. Common-factor PA here uses eigenvalues of that matrix after replacing its diagonal by SMCs. The same SMC rule is applied to the observed and every permuted matrix. This is a precisely defined reduced-matrix diagnostic, not a sequence of ML factor fits, and not a claim that SMCs are known population communalities.

Setting	Declared value
Input	Sample A, all twelve continuous indicators, complete observations
Null generation	Independent within-column permutations without replacement

Setting	Declared value
Matrix	Pearson correlations in each observed/null calculation
Component target	Full unit-diagonal matrix eigenvalues
Factor target	SMC-diagonal reduced-matrix eigenvalues; no iterative extraction
Reference	95th percentile, R quantile type 7
Replications and seed	1,000; 20260924
RNG	Mersenne-Twister / Inversion / Rejection
Failure policy	Stop and investigate; no discarded replicate, smoothing, or replacement seed
Count convention	Leading consecutive dimensions above their matched references

The last convention prevents a late isolated crossing among small or negative reduced roots from being counted as another meaningful dimension. It is declared before reading the curves. Reference choices also matter: a mean comparison and a 95th-percentile comparison are different procedures, not two labels for the same decision rule.

The installed `psych` version's `fa.parallel(sim=FALSE)` option was tested before choosing the implementation. In version 2.5.6 it samples column values **with replacement**, rather than performing the exact permutation specified here. The chapter therefore uses a short explicit permutation loop. This is a reason to inspect the implementation, not an argument that package functions are intrinsically unreliable. The official documentation is useful for identifying the different targets and options.

```
# Compact component version; the companion adds the matched SMC target.
set.seed(20260924)
null_roots <- replicate(1000, {
  X0 <- vapply(X, function(v) sample(v, replace = FALSE),
               numeric(nrow(X)))
  eigen(cor(X0), symmetric = TRUE, only.values = TRUE)$values
})
reference_95 <- apply(null_roots, 1, quantile, probs = .95, type = 7)
```

This code exposes the resampling rule instead of hiding it behind a function name. It is illustrative; the canonical companion also checks positive definiteness and computes the two targets in the same loop. There were zero failures among the 1,000 reference replications. No null matrix was repaired and no replicate was dropped from a quantile.

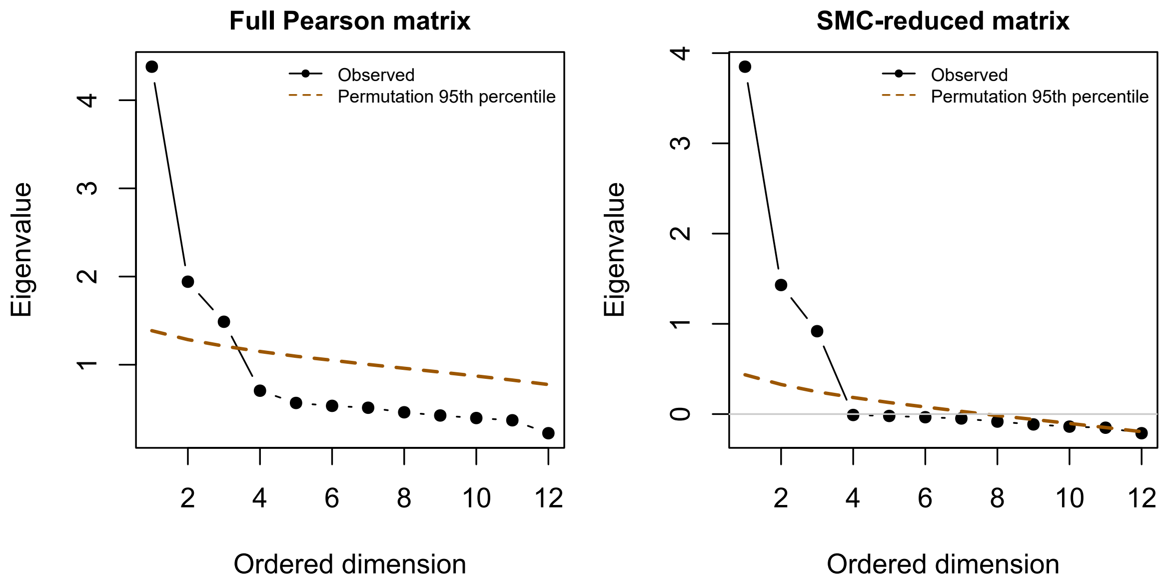

Figure 3: Scree and parallel-analysis evidence. Each observed sequence is compared only with its matching permutation reference.

Table 6

Leading observed and null-reference eigenvalues; full and reduced targets are not pooled.

Dimension	Component	Component 95	Factor	Factor 95
1	4.380	1.386	3.849	0.436
2	1.942	1.285	1.430	0.328
3	1.487	1.211	0.918	0.247
4	0.705	1.150	-0.010	0.184
5	0.566	1.096	-0.021	0.128

Both comparisons retain a leading run of three dimensions. For the third component root, 1.487 exceeds 1.211; for the fourth, 0.705 does not exceed 1.150. The reduced-matrix comparison reaches the same count by a different set of numerical comparisons. Agreement makes the candidate less dependent on this particular diagonal choice, but does not turn the procedure into a proof that three substantive attributes exist.

PA addresses a matched null reference, not every plausible alternative population. Factor correlations, weak factors, minor dimensions, sample size, and the chosen root algorithm can change its performance. It should be read alongside content, neighboring solutions, and residuals.

3.5 A wider family of retention rules

Retention is the decision an exploratory analysis most often gets wrong, and the familiar heuristics are among the weaker available options. Auerswald & Moshagen (2019) compared the main extraction criteria across realistic conditions - correlated factors, unequal loadings, minor factors, non-normal responses, varying sample sizes - and found the eigenvalue-greater-than-one rule and the unaided scree plot performing poorly enough that neither should carry a retention decision alone. Several better-behaved alternatives exist, and they are worth knowing not as a longer checklist but because each states a different reference, and knowing the reference is what lets a disagreement be interpreted.

Parallel analysis is a family, not a method. Two choices generate four distinct procedures. The roots may come from the full Pearson matrix, as in a component analysis, or from a reduced matrix with communality estimates on the diagonal, closer to the common-factor model. The null reference may be the mean of the simulated roots or their 95th percentile. Across Auerswald & Moshagen (2019)'s conditions the component-based versions generally outperformed the reduced-matrix versions, and the mean reference tends to retain more dimensions than the percentile reference. All four share one structural weakness: the reference data are uncorrelated by construction, so after a strong first factor has absorbed much of the observed covariance, the comparison for the second and later roots is no longer like for like, and a genuine second dimension can be missed.

The empirical Kaiser criterion repairs the Kaiser–Guttman rule rather than discarding it (Braeken & Assen, 2017). Under the null model the sample eigenvalues of uncorrelated variables follow a Marchenko–Pastur distribution whose upper bound is $(1+\sqrt{p/n})^2$; this is the reference for the first root. Later roots are corrected for the variance already accounted for, and the reference is never allowed below one:

$$\mathrm{EKC}_j = \max\left\{(1+\sqrt{p/n})^2\,\frac{p-\sum_{i<j} d_i}{p-j+1},\ 1\right\}. \tag{12}$$

With $p = 12$ and $n = 360$ the first reference is 1.398, appreciably above one: a leading root of 1.1 would satisfy Kaiser–Guttman and fail the EKC. From the second root onward the correction term falls below one in this record, so the floor applies and the remaining references equal one. The criterion is therefore stricter than Kaiser–Guttman at the first root and identical afterwards - here, on this data, which is exactly the sort of detail that a printed count conceals.

Sequential model tests use the exact-fit statistic rather than an eigenvalue. Fit a one-factor model; if its test is significant, fit two, and continue until a test is not significant. The logic is clean, but it carries a caveat worth stating: once the number of factors exceeds the true number, the regularity conditions behind the χ^2 reference fail, so the sequence is only trustworthy while it is still under-fitting. An earlier test that was significant by chance therefore propagates into further over-extraction rather than being corrected.

Two further methods appear in the comparison literature and in the companion utilities. The Hull method chooses the count at which the balance between fit and number of parameters stops improving, borrowing the convex-hull idea from model selection (Lorenzo-Seva et al., 2011). Comparison data generates reference datasets that *reproduce* the observed correlation matrix under a known factorial structure of $1, 2, 3, \ldots$ factors, rather than under independence, and retains the smallest count whose reference data resemble the observed data (Ruscio & Roche, 2012). Both address the same weakness in parallel analysis: an uncorrelated null is not the relevant comparison once one factor is granted.

Table 7 applies seven of these rules to the present record, using the same permutation replications throughout so that the comparisons differ only in the stated respect.

Table 7

Seven retention rules on the same development record. What separates them is the reference each uses, not the data.

Rule	Reference	Retained
Kaiser-Guttman	Root exceeds 1	3
Empirical Kaiser criterion	Root exceeds a sample-adjusted bound, floored at 1	3
PA, component roots, mean	Root exceeds the mean null root	3
PA, component roots, 95th	Root exceeds the 95th-percentile null root	3
PA, reduced roots, mean	Reduced root exceeds the mean null reduced root	3
PA, reduced roots, 95th	Reduced root exceeds the 95th-percentile null reduced root	3
Sequential chi-square tests	First non-significant exact-fit test	3

Every rule returns three. That agreement is a fact about this record, not a general property of the methods, and it is what a well-behaved simulated population with three clearly separated factors is expected to produce. Real records disagree often, which is why Auerswald & Moshagen (2019) recommend **combination rules** rather than any single criterion: when sequential model tests agree with either the 95th-percentile component parallel analysis or the empirical Kaiser criterion, the pair identified the correct count with high consistency across their conditions. Sequential tests perform well with highly correlated factors but over-extract in the presence of minor factors and under non-normality; the other two are robust and do not over-extract, which is why they pair well.

The value of a combination rule is not that it counts votes. It is that disagreement is informative: when these criteria diverge, the dimensions are genuinely hard to detect in the available evidence, and the appropriate response is usually more data or a narrower claim rather than a casting vote. A majority vote among heuristics cannot replace explaining what each diagnostic is comparing.

4 Estimate neighboring dimensional solutions

4.1 Count effective parameters, not every printed loading

For p indicators, the symmetric covariance matrix supplies $u = p(p+1)/2$ distinct second moments. An exploratory model with m orthonormal factors appears to contain pm loadings and p uniquenesses. However, orthogonal rotations leave its implied covariance unchanged. The $m(m-1)/2$ rotational degrees of freedom do not add independently recoverable covariance parameters. The effective count is therefore

$$t_m = pm + p - \frac{m(m-1)}{2}, \qquad df_m = \frac{p(p+1)}{2} - t_m = \frac{(p-m)^2 - p - m}{2}.$$

This is a dimension count for the unrestricted common-factor covariance model at regular points. An oblique rotation does not add factor correlations to this count: it re-expresses the same fitted covariance. Counting every oblique loading, every factor correlation, and every uniqueness as independently identified would count the same freedom more than once.

Table 8
Covariance-moment accounting for 12 indicators.

Factors	Moments	Effective parameters	Candidate df
1	78	24	54
2	78	35	43
3	78	45	33
4	78	54	24
5	78	62	16

The nonnegative-df condition also yields the familiar upper counting bound

$$m \leq \frac{2p + 1 - \sqrt{8p+1}}{2}.$$

For $p = 12$, the right side is about 7.58, so the integer count allows at most seven factors. This bound is not a recommendation to extract seven, a count of positive eigenvalues, or a proof of identification. Particular loading patterns can be deficient, and a solution can be weakly determined even when the generic count looks comfortable. Chapter 3 examines these distinctions directly.

4.2 Fit one through five; inspect two, three, and four closely

We estimate normal-theory ML common-factor models for one through five factors. Each fit uses the same Pearson matrix, $n = 360$, SMC starts, and the same uniqueness-bound convention. The primary extraction is unrotated; rotation is a subsequent coordinate choice. The printed code shows the essential calls, while the shared script records convergence checks and full settings.

```
fit3 <- psych::fa(R, nfactors = 3, n.obs = nrow(x),
                  fm = "ml", rotate = "none", SMC = TRUE,
                  min.err = 1e-8, max.iter = 1000,
                  smooth = FALSE, scores = "none")
A <- unclass(fit3$loadings)
```

The implemented analysis also checks the ML objective using `stats::factanal()` with five starting values. Agreement of independently computed objectives is a useful numerical check, not proof that the scientific model is correct. A numerical algorithm can converge perfectly to an inadequate model.

Table 9
Neighboring ML solutions. RMS uses distinct off-diagonal correlation residuals.

Factors	df	F ML	RMS offdiag	Max residual	Min uniqueness	Boundary
1	54	2.063	0.146	0.497	0.445	FALSE
2	43	0.797	0.086	0.321	0.245	FALSE
3	33	0.089	0.015	0.034	0.202	FALSE
4	24	0.050	0.013	0.030	0.005	TRUE
5	16	0.022	0.009	0.027	0.005	TRUE

The two-factor solution leaves a largest absolute residual correlation of 0.321, compared with 0.034 for three factors. That is a substantial change in the covariance reconstruction, not merely an additional column of interpretable words. Adding a fourth factor reduces the discrepancy further, but its smallest uniqueness is approximately .005, at the numerical lower bound. The fifth-factor solution does the same. A bound prevents an optimizer from reporting a negative uniqueness; it does not make that boundary estimate substantively credible.

An interior uniqueness of .20 and a bound-constrained uniqueness of .005 are qualitatively different evidence. The latter can signal too many factors, a highly redundant pair, sampling fluctuation, or a misspecified common-factor model. Inspect its indicator and neighboring pattern before assigning a new construct name. Do not silently omit the bound from the report or describe it as evidence of nearly perfect measurement.

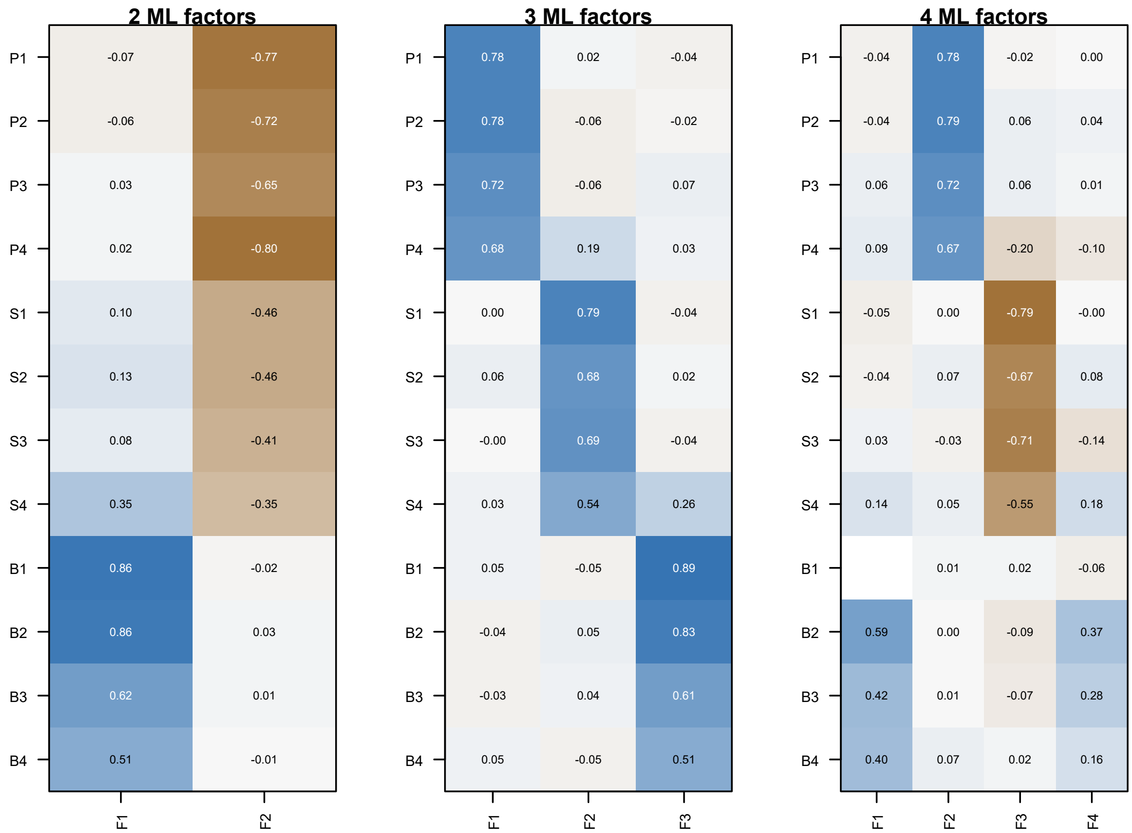

Figure 4: Oblimin patterns for neighboring dimensions. Column signs and order are arbitrary across different factor counts; compare item groupings, not column numbers.

Read this display in two directions. Down a column, ask which items define a coherent factor. Across a row, ask whether an item has a stable primary relation or is distributed across factors. A smaller model must represent more covariance using fewer common directions; it may merge domains or leave patterned residuals. A larger model can separate narrow item-specific covariance that does not justify an additional broad construct. In neither case is a short label an adequate diagnostic.

4.3 An exact-fit test is one piece of evidence

For reference, `psych` reports a corrected ML statistic of the form

$$T_m = \left[n - 1 - \frac{2p+5}{6} - \frac{2m}{3}\right] \widehat{F}_{\mathrm{ML}}.$$

Under the exact factor model and suitable regularity conditions, its reference is approximately $\chi^2_{df_m}$. For the three-factor fit, $T_3 = 31.327$ with 33 df, giving $p = 0.551$. This does not establish that the factor interpretation is true; it says that this discrepancy is not unusual under the exact-model reference used by the test. Power and misspecification size matter. A nonsignificant test cannot establish equivalence or rule out consequential departures.

The fourth- and fifth-factor solutions reach a variance boundary, where ordinary interior-point chi-square arguments need additional care. Their large conventional p values should not be used to endorse extra factors. Nor should one subtract any two displayed factor tests and automatically declare a regular nested likelihood-ratio test: changes of factor count involve singular or boundary configurations under the smaller model.

Chapter 3 uses raw-data ML with an explicit normal-likelihood convention. Its test multiplier is not this small-sample EFA correction. Comparability of estimation families does not mean every package reports an identical finite-sample test statistic.

5 Resolve the orientation by rotation

5.1 What rotation changes, and what it preserves

An unrotated solution is convenient for estimation but often inconvenient for interpretation. Suppose the extracted common covariance is $\mathbf{A}\mathbf{A}^\top$. For an orthogonal matrix $\mathbf{Q}$, with $\mathbf{Q}^\top\mathbf{Q} = \mathbf{I}$,

$$\mathbf{P} = \mathbf{A}\mathbf{Q} \quad \Longrightarrow \quad \mathbf{P}\mathbf{P}^\top = \mathbf{A}\mathbf{A}^\top.$$

The rows of $\mathbf{P}$ locate the indicators in a rotated coordinate system. The reproduced covariance, communalities, uniquenesses, and extraction discrepancy are unchanged. The individual loading coefficients change because they refer to different axes.

The general nonsingular transformation requires a paired change. Let $\eta^* = \mathbf{T}\eta$, where the original factor covariance is $\mathbf{I}$. Then

$$\mathbf{P} = \mathbf{A}\mathbf{T}^{-1}, \qquad \Phi^* = \mathbf{T}\mathbf{T}^\top,$$

and

$$\begin{aligned}\mathbf{P}\Phi^*\mathbf{P}^\top &= \mathbf{A}\mathbf{T}^{-1}\mathbf{T}\mathbf{T}^\top(\mathbf{T}^{-1})^\top\mathbf{A}^\top \\ &= \mathbf{A}\mathbf{A}^\top.\end{aligned}$$

If the transformed factor variances are subsequently rescaled to one, the loading columns must be rescaled correspondingly. Multiplying the loadings by an arbitrary matrix while leaving

Φ unchanged is not an equivalent oblique rotation. This inverse-paired transformation is the same coordinate principle introduced in Chapter 1.

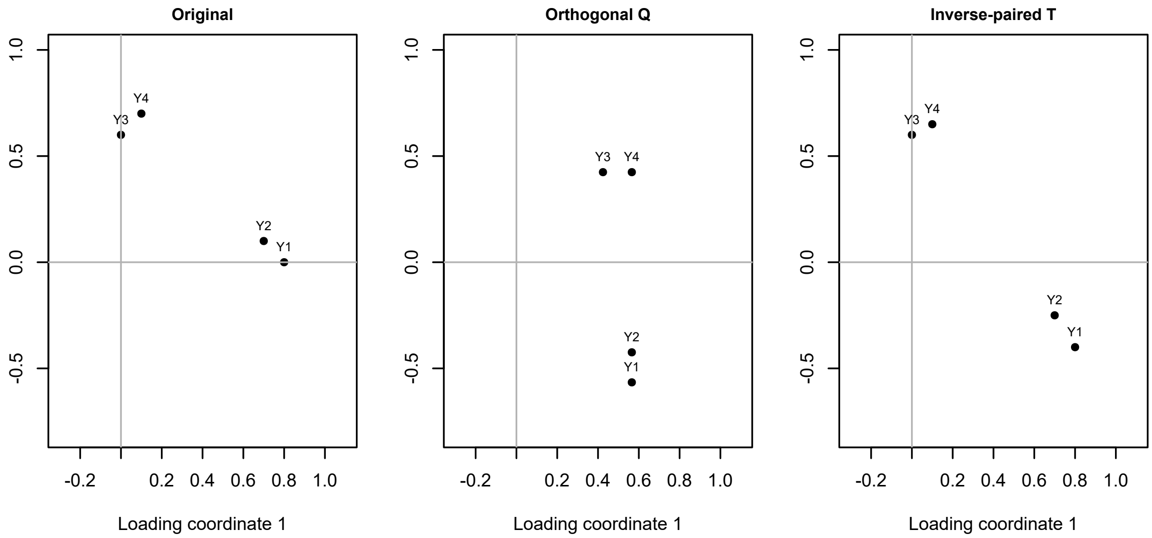

Figure 5: Coordinate changes in a two-dimensional illustration. These are illustrative loadings, not a second empirical data set.

5.2 Orthogonal and oblique simple structure

Simple structure favors interpretable differentiation: relatively few substantial relations per item and identifiable groups of indicators. It is an orientation criterion, not a new criterion of fit to the observed covariance matrix. Rotation does not improve F_{ML}.

Varimax is an orthogonal criterion. In an unnormalized version it maximizes, across columns, the dispersion of squared loadings:

$$V(\mathbf{P}) = \sum_{a=1}^{m} \left\{ \frac{1}{p} \sum_{j=1}^{p} p_{ja}^4 - \left(\frac{1}{p} \sum_{j=1}^{p} p_{ja}^2 \right)^2 \right\}.$$

A column with some larger and some smaller squared loadings has a higher criterion than a column whose squared loadings are all similar. The orthogonality constraint remains in force; a clean-looking varimax pattern does not establish that the substantive attributes are unrelated.

Oblique criteria permit factor correlations while seeking a simpler pattern. Here the primary rotation is direct oblimin with $\gamma = 0$ (quartimin), no Kaiser normalization, convergence tolerance 10^{-7}, up to 10,000 iterations, and 20 random starts. At $\gamma = 0$, the criterion discourages a row from having large squared coefficients in several columns. It does not set the factor correlations to zero.

```
set.seed(20260928)
rot3 <- GPArotation::oblimin(A, gam = 0, normalize = FALSE,
                              eps = 1e-7, maxit = 10000,
                              randomStarts = 20)
pattern <- unclass(rot3$loadings)
factor_correlations <- rot3$Phi
```

Kaiser normalization, when requested, divides each row by its communality-based length before optimizing and restores its scale afterward. That changes the criterion's relative weighting of indicators. It is not the same operation as standardizing the original observed variables. Because implementation details can affect the chosen orientation, the rotation name alone is not a sufficient specification.

Multiple starts reduce dependence on a single starting orientation but do not prove that a global optimum was found. Signs and column order are also arbitrary. Our reported solution aligns columns with the first three content anchors in each domain and makes their summed primary loadings positive. This is a labelling convention applied after fitting; it is not an extra model restriction.

5.3 Why the oblique solution is primary here

Planning, strategy adjustment, and belonging need not be independent. The oblimin solution estimates the following factor correlations:

Table 10
Factor correlation matrix in the aligned three-factor oblimin solution.

Factor	Planning	Strategy	Belonging
Planning	1.000	0.464	0.312
Strategy	0.464	1.000	0.336
Belonging	0.312	0.336	1.000

These are estimated relations between the chosen common-factor coordinates. They are not causal paths, observed scale correlations, or proof that the same orientation would be uniquely recovered in another sample. Varimax provides a useful comparison precisely because it represents the same common covariance with orthogonal coordinates. The maximum absolute difference in reproduced common covariance between the reported oblimin and varimax solutions is 4.4e-16. The near-zero value verifies invariance; it does not make their loading tables interchangeable.

6 Read an oblique factor system

6.1 Pattern coefficients and structure correlations

With standardized indicators and unit-variance factors, the pattern matrix $\mathbf{P}$ contains the coefficients in

$$\mathbf{z}_i = \mathbf{P}\eta_i + \epsilon_i.$$

Each row is a simultaneous linear decomposition. A coefficient describes the relation of that factor to the indicator conditional on the other factors in this representation. The structure matrix instead gives the zero-order indicator–factor correlations:

$$\mathbf{C} = \mathrm{Cov}(\mathbf{z}, \eta) = \mathbf{P}\Phi.$$

Since both sides have unit marginal variances, those covariances are correlations. In an orthogonal solution $\Phi = \mathbf{I}$, so pattern and structure coincide. In an oblique solution they generally do not.

Table 11

All pattern coefficients. Small entries remain visible; none were suppressed at .30.

Item	Planning	Strategy	Belonging
P1	0.783	0.018	-0.040
P2	0.784	-0.056	-0.021
P3	0.720	-0.062	0.066
P4	0.679	0.188	0.033
S1	0.000	0.794	-0.037
S2	0.057	0.677	0.016
S3	-0.002	0.685	-0.036
S4	0.030	0.536	0.261
B1	0.045	-0.047	0.894
B2	-0.044	0.055	0.829
B3	-0.031	0.041	0.610
B4	0.046	-0.047	0.515

Table 12
Structure correlations for the same orientation.

Item	Planning	Strategy	Belonging
P1	0.779	0.368	0.210
P2	0.751	0.300	0.205
P3	0.712	0.294	0.270
P4	0.777	0.514	0.308
S1	0.357	0.781	0.229
S2	0.376	0.709	0.261
S3	0.305	0.672	0.193
S4	0.361	0.638	0.451
B1	0.303	0.274	0.892
B2	0.240	0.313	0.834
B3	0.179	0.232	0.614
B4	0.185	0.147	0.513

For P4, the Strategy structure correlation is

$$c_{\mathrm{P4,S}} = p_{\mathrm{P4,P}}\phi_{\mathrm{PS}} + p_{\mathrm{P4,S}} + p_{\mathrm{P4,B}}\phi_{\mathrm{BS}}.$$

Substituting the unrounded estimates gives 0.514. The direct Strategy pattern coefficient is only 0.188. Much of the zero-order relation reflects P4's Planning loading together with the Planning–Strategy association. Reading the structure matrix as if every entry were a separate direct loading would therefore exaggerate cross-loading complexity.

Conversely, dismissing every secondary pattern coefficient below .30 would conceal the two content-relevant relations under consideration. P4 concerns revising a plan when a method fails; S4 concerns using others to improve a learning strategy. Their secondary coefficients, 0.188 and 0.261, are evidence to evaluate alongside content, orientation sensitivity, and later independent estimation. They are not established nonzero population coefficients merely because we can give them plausible names.

6.2 Communality is a quadratic form

For indicator j, let $\mathbf{p}_j^\top$ be row j of the pattern matrix. Its common variance is

$$h_j^2 = \mathbf{p}_j^\top \Phi \mathbf{p}_j, \qquad u_j^2 = 1 - h_j^2.$$

For three standardized factors this expands to

$$h_j^2 = p_{j1}^2 + p_{j2}^2 + p_{j3}^2 + 2p_{j1}p_{j2}\phi_{12} + 2p_{j1}p_{j3}\phi_{13} + 2p_{j2}p_{j3}\phi_{23}.$$

The cross-products are part of the common variance. Summing squared pattern coefficients alone drops them. For P4, the row sum of squared pattern coefficients is 0.498, whereas the communality is 0.635. Neither number should be labelled the variance uniquely explained by Planning.

Table 13

Common and unique variance in the standardized metric. Row sums of squared pattern coefficients are included only to expose the oblique distinction.

Item	Pattern SS	Communality	Uniqueness
P1	0.616	0.609	0.391
P2	0.618	0.568	0.432
P3	0.527	0.512	0.488
P4	0.498	0.635	0.365
S1	0.631	0.612	0.388
S2	0.462	0.505	0.495
S3	0.471	0.453	0.547
S4	0.356	0.470	0.530
B1	0.803	0.798	0.202
B2	0.693	0.698	0.302
B3	0.374	0.379	0.621
B4	0.269	0.266	0.734

The column sums of squared loadings reported by software are useful descriptions of an orientation, but correlated factors do not provide a unique additive allocation of common variance to separate constructs. One cannot turn overlapping factor relations into mutually exclusive percentages simply by dividing each column sum by p.

A low communality can mean that an indicator is weakly represented by this common-factor system. It does not by itself show poor wording or justify removal. An indicator may intentionally cover content absent from the other items. Removing it can make the covariance model cleaner while narrowing the construct. A high communality can likewise arise from redundancy rather than desirable breadth.

6.3 Reconstruct a pair, then the whole matrix

Under diagonal uniqueness, the reproduced correlation of distinct indicators j and k is

$$\hat{r}_{jk} = \hat{\mathbf{p}}_j^\top \widehat{\Phi} \hat{\mathbf{p}}_k.$$

All primary and secondary coefficients participate. For P4 and S4, the observed correlation is 0.368 and the reproduced correlation is 0.380. The residual, defined as observed minus reproduced, is -0.012. Its sign is interpretable only after the subtraction convention is stated.

```
common <- pattern %*% factor_correlations %*% t(pattern)
communality <- diag(common)
reproduced <- common + diag(fit3$uniquenesses)
residual <- R - reproduced
```

This short calculation is more informative than memorizing the names of several output tables. It links loadings, factor correlations, communalities, uniquenesses, and residuals to one covariance equation. When a table cannot be reconstructed from that equation, first check the standardization and rotation conventions rather than inventing a substantive explanation.

6.4 What is uncertain about this interpretation?

Sampling affects extraction, factor count, orientation, and the apparent stability of item assignments. A coefficient reported to three decimals is not thereby precise to three decimals. The comparisons below examine specified analytic alternatives on one sample; they do not provide a sampling confidence interval. Bootstrap analysis would additionally require re-estimation, alignment, and a decision about whether the factor count is held fixed. Unaligned bootstrap columns cannot sensibly be averaged.

Factor scores are another distinct object. EFA estimates a population covariance representation; individual latent values are not observed merely because loadings have been estimated. Alternative scoring rules can produce different scores under the same factor model. Chapter 4 develops this issue after measurement-model evaluation.

7 Check reconstruction and analytic sensitivity

7.1 Examine local residuals, not only an average

Define the correlation residual matrix

$$\mathbf{E} = \mathbf{R} - \widehat{\mathbf{R}}, \qquad \mathrm{RMS}_{\mathrm{off}} = \sqrt{\frac{2}{p(p-1)} \sum_{j<k} e_{jk}^2}.$$

The off-diagonal average deliberately excludes variances. It is not automatically identical to every software statistic called RMSR or SRMR. Its definition here is transparent and reproducible. The largest absolute residual supplies different information: one problematic pair can be concealed by averaging many small residuals.

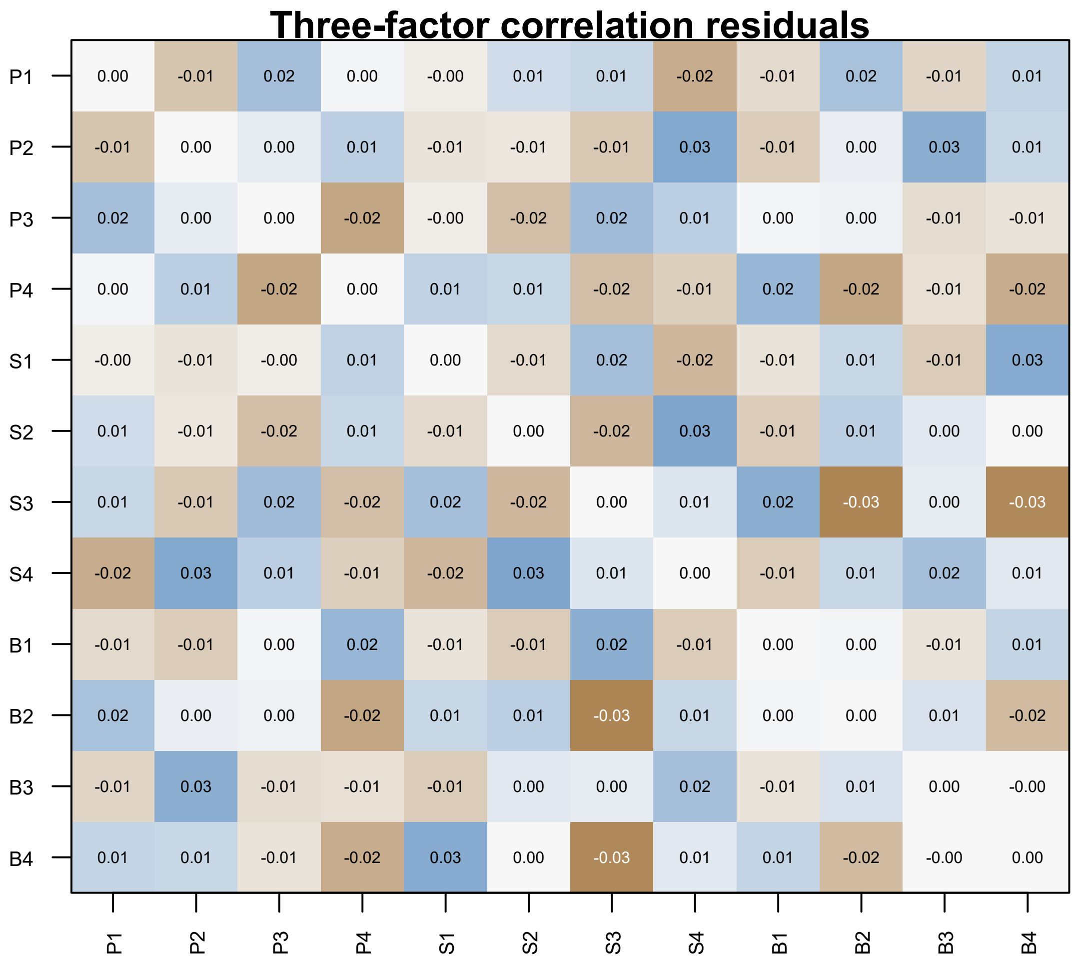

Figure 6: Observed minus reproduced correlations for the three-factor ML solution. The scale is centered on zero.

Table 14
Eight largest absolute off-diagonal residuals, with observed and reproduced values.

Item1	Item2	Observed	Reproduced	Residual
B2	S3	0.150	0.184	-0.034
B4	S3	0.049	0.082	-0.033
S4	S2	0.487	0.459	0.027
S4	P2	0.264	0.237	0.027
B4	S1	0.123	0.098	0.026
B3	P2	0.139	0.114	0.025
B1	S3	0.180	0.155	0.025
B2	P4	0.225	0.249	-0.024

Ask whether residuals concentrate within a domain, between two domains, or around a particular item. Those patterns suggest different questions about omitted common dimensions, secondary loadings, and local residual relations. A residual is evidence of unreproduced covariance, not a label for its cause. An unrestricted exploratory loading pattern can also absorb a local relation into its common part. Small EFA residuals therefore do not certify that every later simple-structure CFA restriction is safe.

7.2 KMO and Bartlett answer preliminary questions

Let a_{jk} be the partial correlation between indicators j and k controlling the remaining indicators. A common overall KMO statistic is

$$\text{KMO} = \frac{\sum_{j<k} r_{jk}^2}{\sum_{j<k} r_{jk}^2 + \sum_{j<k} a_{jk}^2}.$$

The sample value is 0.843. It is larger when the squared partial correlations are small relative to the squared zero-order correlations. That is a useful compact description, but it neither selects the number of factors nor establishes the meaning of any factor. It is also not a universal minimum-sample-size calculation.

Bartlett's sphericity test addresses $H_0 : \mathbf{R}_{\text{population}} = \mathbf{I}$. A familiar normal-theory approximation is

$$T_B = -\left[n - 1 - \frac{2p+5}{6}\right] \log|\mathbf{R}|, \qquad df_B = \frac{p(p-1)}{2}.$$

Here $T_B = 1675.5$ on 66 df. Rejecting an identity correlation matrix says that the indicators are not mutually uncorrelated in the population under the test assumptions. It does not show that a particular low-dimensional factor model fits. A matrix with complicated local dependencies can readily reject sphericity.

Use these summaries to characterize the input, not as entrance tickets that replace inspecting the correlation matrix, positive definiteness, indicator content, and the fitted model.

7.3 Change one analytic choice at a time

Two bounded sensitivity analyses use the same observations, variables, and Pearson matrix. First, keep ML extraction and use oblique geomin rotation with $\delta = .01$, no normalization, and 20 starts. Second, keep three factors and oblimin but replace ML extraction by minres. The first changes the orientation criterion; the second changes how the covariance discrepancy is weighted.

Signs and factor order are aligned before comparing coefficients. For loading columns $\mathbf{a}$ and $\mathbf{b}$, Tucker's congruence coefficient is

$$c(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\mathsf{T}\mathbf{b}}{\sqrt{\mathbf{a}^\mathsf{T}\mathbf{a}}\sqrt{\mathbf{b}^\mathsf{T}\mathbf{b}}}.$$

It compares the directions of two coefficient vectors, not the correlation between participant scores. Near-one congruence can coexist with an important difference for one item, so also inspect the largest coefficient difference and the relevant rows.

Table 15

Aligned sensitivity comparisons against primary ML oblimin. These are same-sample analytic checks, not independent replications.

Analysis	Max pattern difference	Max Phi difference	Min congruence
ML oblimin	0.000	0.000	1.000
ML geomin	0.029	0.022	0.999
Minres oblimin	0.011	0.001	1.000

The largest pattern change under geomin is 0.029; under minres it is 0.011. The principal item groupings and the two content-relevant secondary relations persist. This supports carrying a specific candidate forward without claiming that all reasonable analyses have been exhausted. Changing the item set, sample, distributional assumptions, or factor count would be a different sensitivity question.

The defensible conclusion is deliberately narrower than "the scale is validated." Three correlated dimensions summarize Sample A better than two, the main orientation is interpretable and stable across these limited checks, and the larger fitted solutions encounter boundaries. These observations justify a candidate to test elsewhere.

8 Summary and reading bridge

EFA is a sequence of linked decisions. The input matrix establishes the observed metric. Extraction states which covariance structure is being approximated and how discrepancies are weighted. Retention selects a plausible dimension using matched reference comparisons, content, and neighboring solutions. Rotation chooses interpretable coordinates without changing the fitted covariance. Interpretation then requires pattern, structure, factor correlations, and the quadratic-form communality to be read together.

The Sample A record supports a three-factor candidate, not a final measurement verdict. Both declared parallel analyses retain three dimensions; two factors leave appreciably larger residuals; four and five reach uniqueness bounds. The primary oblique pattern and limited same-sample sensitivity checks support three content-defined domains - Planning, Strategy and Belonging - together with two secondary loadings that are retained on content grounds rather than by a numerical cutoff: P4 on Strategy and S4 on Belonging. All twelve indicators are kept, in the codebook order, and every other cross-loading and every residual covariance would be fixed at zero in a restricted version of this model. That specification, and the stricter simple-structure alternative that fixes both secondary loadings to zero, are what a confirmatory analysis on unused data would have to test.

A compact report could read:

> We analyzed Pearson correlations for 12 complete continuous indicators in a simulated development sample of 360 cases. Matched 95th-percentile permutation references for full and SMC-reduced eigenvalues both retained three dimensions. Neighboring ML models and residual inspection supported examining a three-factor representation; four- and five-factor fits reached uniqueness bounds. Direct oblimin with stated settings produced interpretable correlated domains. Geomin and minres checks gave closely aligned patterns. A 12-indicator CFA candidate with two content-motivated secondary loadings was frozen for independent-sample estimation; this development analysis is not itself confirmation.

For mathematical background, connect the common-factor representation and rotational indeterminacy to Mulaik (2010) and Bartholomew & Knott (1999). Read Horn (1965) for the parallel-analysis idea and Buja & Eyuboglu (1992) for the permutation perspective and qualifications. The historical oblique rotation literature includes Jennrich & Sampson (1966). Software documentation is indispensable for implementation details, especially distinctions between simulation, resampling, and permutation.

Chapter 3 now changes the question: **given the frozen zero, free, and equality restrictions, can a CFA be specified, identified, and estimated coherently?** An exploratory orientation becomes a confirmatory model only when its restrictions and analytic history are made explicit.

Exercises

The following exercises are answer-free. Use the printed tables where possible; code is optional unless a prompt explicitly asks for it.

1. **Foundational – identify the target.** An analyst extracts principal components from $\mathbf{R}$ and calls the discarded component variance "measurement error." Explain the missing common-factor assumptions. For $\mathbf{R} = \begin{bmatrix} 1 & .6 \\ .6 & 1 \end{bmatrix}$, obtain its eigenvalues and the variance represented by the first component.
2. **Foundational – update a diagonal.** For the three-variable matrix $\mathbf{R} = \begin{bmatrix} 1 & .5 & .4 \\ .5 & 1 & .3 \\ .4 & .3 & 1 \end{bmatrix}$, calculate the SMCs using $1 - 1/(\mathbf{R}^{-1})_{jj}$. Form the reduced matrix and describe one principal-axis update. Explain why a negative reduced-matrix eigenvalue is not automatically a negative variance.
3. **Foundational – audit the count.** Derive the effective parameter counts and candidate df for $p = 8$, $m = 1, 2, 3$. Explain the rotational subtraction and why a nonnegative count does not prove identification.
4. **Foundational – match the null.** At ranks 1–4, observed roots are $3.8, 1.5, .9, .2$, and matching permutation 95th percentiles are $.5, .3, .2, .1$. Determine the leading crossing count. Explain why this table cannot be interpreted without knowing which matrix generated the roots, and why a full-matrix reference cannot replace a reduced one.
5. **Foundational – verify a coordinate change.** Let $\mathbf{A} = \begin{bmatrix} .8 & .1 \\ .2 & .7 \\ .5 & .4 \end{bmatrix}$ and $\mathbf{T} = \begin{bmatrix} 1 & .3 \\ 0 & 1 \end{bmatrix}$. Calculate $\mathbf{P} = \mathbf{A}\mathbf{T}^{-1}$ and $\Phi^* = \mathbf{T}\mathbf{T}^\mathsf{T}$, and verify the common-covariance identity. Is Φ^* already a correlation matrix?
6. **Foundational – read an oblique row.** For pattern row $(.6, .3)$ and factor correlation $.4$, compute the two structure correlations, communality, and uniqueness of a standardized indicator. Explain why the sum of squared pattern coefficients gives a different result.
7. **Foundational – reconstruct a pair.** Take pattern rows $(.7, .1)$ and $(.2, .6)$ with factor correlation $.35$. Compute their reproduced correlation. If the observed correlation is $.45$, calculate the signed residual and explain what it does and does not diagnose.
8. **Integrative – compare dimensions.** Using the one-to-five-factor table and neighboring-pattern figure, recommend a dimension for further study. Discuss the uniqueness boundary and explain why neither the smallest discrepancy nor the largest conventional p value is a sufficient decision rule.
9. **Integrative – evaluate an item.** Contrast P4 and B4 using wording, pattern, structure, and communalities. Propose what additional evidence would be needed before deleting either. Do not use a loading cutoff as the complete rationale.
10. **Integrative – audit sensitivity.** Explain what is held constant in the ML/geomin and minres/oblimin comparisons. Why must factor order and sign be aligned? Identify one

question these comparisons cannot answer about sampling uncertainty.

11. **Integrative – reproduce the contract.** Write plain R code to read Sample A, select the codebook item order, check completeness, and compute the Pearson matrix. Then specify how to generate an exact column-permutation reference with a fixed seed. Explain why resampling with replacement is a different reference procedure.
12. **Integrative – state the handoff.** Write the free/fixed relation ledger for two restricted models: the three-factor candidate that keeps the P4 and S4 secondary loadings free, and the stricter version that fixes both at zero. Give the free-parameter count and candidate df for each against the 78 distinct moments of twelve indicators. Then identify which evidence could and could not be used to choose between them, and explain what changes if a model is revised after inspecting a second sample.

Appendix A: SLE data dictionary and sample manifest

The downloadable file sle_continuous.csv contains 1,080 simulated records and 14 columns: two identifiers and twelve continuous indicators. The codebook and manifest are printed here so that no separate metadata file is needed. The three samples have been combined for distribution only; no observations have been pooled for analysis, regenerated, rounded, standardized, or reassigned.

Fields and common coding

Table 16
CSV fields and coding rules. Neither sample nor id is a factor indicator.

Field	Meaning
sample	A = development; B = confirmation; C = revalidation.
id	Unique synthetic record ID; the first letter agrees with sample.
P1-P4	Proposed Planning domain.
S1-S4	Proposed Strategy domain.
B1-B4	Proposed Belonging domain.
All twelve indicators	Continuous, unbounded process-index points.
Direction	Higher values mean more of the stated attribute; no reverse coding.
Missing values	NA is the missing-value code; the supplied records have no missing observations.
Response format	Hypothetical continuous indices, not single ordered-category responses, percentages, or counts.

Item codebook

The wording gives illustrative content, not evidence that a questionnaire was administered or validated. Proposed domains precede EFA and do not guarantee the recovered factor structure.

Table 17
Complete SLE item definitions and illustrative wording; the common coding rules above apply to every row.

Item	Domain	Definition	Wording
P1	Planning	Planning-routine consistency	I establish a study plan before beginning a task.
P2	Planning	Priority-setting clarity	I identify which study tasks deserve attention first.
P3	Planning	Progress-monitoring regularity	I compare my progress with the plan I made.
P4	Planning	Adaptive planning	I revise my plan when a study method is not working.
S1	Strategy	Strategy-adjustment flexibility	I change my learning strategy when understanding is weak.
S2	Strategy	Method-selection deliberateness	I select a learning method suited to the task.
S3	Strategy	Reflective revision	I revise my explanation after checking my understanding.
S4	Strategy	Socially supported strategy change	I use discussion with others to improve my learning method.
B1	Belonging	Peer acceptance	I feel accepted by other students.
B2	Belonging	Peer participation	I feel able to join other students in learning activities.
B3	Belonging	Classroom voice	I feel able to express my views in class.
B4	Belonging	Institutional connection	I feel connected to the university as a whole.

The Planning and Strategy indicators are hypothetical continuous process indices; Belonging indicators are hypothetical continuous experience indices. The following qualifications make overlapping content explicit. They do not disclose an estimated loading or authorize a model change.

Table 18
Indicator-specific response-process qualifications.

Item	Qualification
P4	Combines planning and method adjustment.
S4	Combines learning strategy and social participation.
B1 and B2	Both refer to a shared peer context.

Sample manifest

Table 19

Three independent samples in one file. The first-use rules concern analytical evidence, not physical file accessibility.

Sample	N	IDs	Seed	Permitted first use
A: development	360	A001-A360	20260921	Chapter 2: EFA development.
B: confirmation	360	B001-B360	20260922	Chapter 3: estimate the candidate frozen after A; Chapter 4: evaluate it.
C: revalidation	360	C001-C360	20260923	Chapter 4: retest the revision frozen after B.

Each row belongs to one sample only. The 360-case sample sizes are pedagogical choices, not general adequacy recommendations. Importing the combined file does not authorize earlier use of B or C. Select the eligible sample before descriptive summaries, missing-data decisions, rescaling, correlations, extraction, estimation, or diagnostics. Keep `sample` and `id` out of the indicator matrix.

The records are simulated from the declared population with the seeds in Table 19. The chapter regenerates the development sample in memory each time it renders, so nothing has to be downloaded to reproduce every number printed here; the distributed `sle_continuous.csv` contains the same records for readers who prefer to work from a file. The generation specification was declared before any sample was inspected, and no seed was chosen for its fit results.

Appendix B: Spectral and rotation details

Why a truncated eigendecomposition is a component solution

For a symmetric positive-semidefinite matrix, $\mathbf{R} = \mathbf{V}\mathbf{D}\mathbf{V}^{\mathsf{T}}$ with orthonormal eigenvectors. Retaining the first m eigenpairs gives $\mathbf{R}_m = \mathbf{V}_m\mathbf{D}_m\mathbf{V}_m^{\mathsf{T}}$. Its trace is the sum of the retained eigenvalues. Among unrestricted rank-m approximations, this truncation minimizes squared Frobenius error. The omitted diagonal is not required to be a diagonal residual covariance from a latent-variable measurement model. That is the key reason a convenient low-rank approximation and common-factor estimation answer different questions.

Replacing the diagonal by communality estimates changes the eigenproblem. The reduced matrix need not be positive semidefinite: its diagonal is no longer the full variance of an observed random vector. In a principal-axis iteration, the retained positive roots define a current common part whose diagonal supplies the next communality estimates. Convergence of that iteration and its sensitivity to starts are numerical questions separate from whether the resulting factors make sense.

Other retention summaries are not interchangeable votes

The minimum average partial (MAP) idea progressively removes component directions and examines the remaining partial-correlation structure. It is not the same reference comparison as parallel analysis. Very simple structure (VSS) compares approximations obtained by simplifying loading patterns to a small number of entries per row; it consequently depends on the kind of simplicity one is prepared to impose. Sequential model tests target exact covariance restrictions under their reference conditions. Information criteria trade likelihood against parameter count but also require a carefully specified likelihood and admissible candidate family.

These methods can disagree because they target different features. A useful extension would state each target, its tuning choices, and the implications of disagreement, rather than assemble a majority-vote factor count. Newer eigenvalue corrections likewise require their own derivation and calibration; they should not be introduced merely by renaming the eigenvalue-greater-than-one rule.

Two explicit oblique criteria

Ignoring a positive multiplicative constant, the quartimin objective is

$$Q_{\text{quartimin}}(\mathbf{P}) = \sum_{j=1}^{p} \sum_{a<b} p_{ja}^2 p_{jb}^2.$$

Large coefficients in two columns of the same row increase this objective. A geomin objective takes the form

$$Q_{\text{geomin}}(\mathbf{P}) = \sum_{j=1}^{p} \left[\prod_{a=1}^{m} (p_{ja}^2 + \delta) \right]^{1/m}, \qquad \delta > 0.$$

The small positive δ regularizes behavior near zero. These criteria may have several local optima. Convergence checks, multiple starts, and sensitivity comparisons are therefore part of a complete rotation report. Neither objective supplies a probability that a factor name is correct, and neither changes the fitted common covariance when implemented as an equivalent rotation.

Factor-score indeterminacy in one equation

If factors and indicators are jointly normal, a model-based conditional mean score in the standardized metric is

$$E(\eta \mid \mathbf{z}) = \Phi \mathbf{P}^{\top} \Sigma_z^{-1} \mathbf{z}.$$

The conditional variance generally remains positive. Thus knowing the model parameters does not identify each person's latent value without error. Other scoring criteria lead to other point estimates. This appendix supplies the conceptual warning; Chapter 4 develops scoring and its uncertainty after the CFA model has been evaluated.

References

Auerswald, M., & Moshagen, M. (2019). How to determine the number of factors to retain in exploratory factor analysis: A comparison of extraction methods under realistic conditions. *Psychological Methods*, *24*(4), 468–491. https://doi.org/10.1037/met0000200

Bartholomew, D. J., & Knott, M. (1999). *Latent variable models and factor analysis* (2nd ed.). Oxford University Press.

Braeken, J., & Assen, M. A. L. M. van. (2017). An empirical Kaiser criterion. *Psychological Methods*, *22*(3), 450–466. https://doi.org/10.1037/met0000074

Buja, A., & Eyuboglu, N. (1992). Remarks on parallel analysis. *Multivariate Behavioral Research*, *27*(4), 509–540. https://doi.org/10.1207/s15327906mbr2704_2

Horn, J. L. (1965). A rationale and test for the number of factors in factor analysis. *Psychometrika*, *30*(2), 179–185. https://doi.org/10.1007/BF02289447

Jennrich, R. I., & Sampson, P. F. (1966). Rotation for simple loadings. *Psychometrika*, *31*(3), 313–323. https://doi.org/10.1007/BF02289465

Lorenzo-Seva, U., Timmerman, M. E., & Kiers, H. A. L. (2011). The Hull method for selecting the number of common factors. *Multivariate Behavioral Research*, *46*(2), 340–364. https://doi.org/10.1080/00273171.2011.564527

Mulaik, S. A. (2010). *Foundations of factor analysis* (2nd ed.). Chapman; Hall/CRC.

Ruscio, J., & Roche, B. (2012). Determining the number of factors to retain in an exploratory factor analysis using comparison data of known factorial structure. *Psychological Assessment*, *24*(2), 282–292. https://doi.org/10.1037/a0025697